

SACHA DURAIN

PhD Candidate – Structural Dynamics
& Scientific Machine Learning

CONTACT

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Available from Sep. 2026

CORE SKILLS

Machine & Deep Learning

Supervised/unsupervised ML, vision,
uncertainty quantification, PyTorch,
scikit-learn

Numerical Simulation

Ansys, Abaqus, HyperMesh

Data Analysis

NumPy, pandas, Matplotlib, Plotly

Programming

Python, MATLAB, Bash, HPC, Slurm, Git

LANGUAGES

French – Native
English – C1

RESEARCH OUTPUT

1 published article – MSSP
2 submitted articles – MSSP, JSV
2 conference talks – CSMA, CFM

STRENGTHS

Multidisciplinary profile
Scientific rigor
International teamwork
Autonomy and initiative

INTERESTS

Fitness – 10+ years
Trail running – 5 races

PROFILE

Hybrid background in mechanics and artificial intelligence, bridging experimental testing, finite element simulation, signal processing, and machine learning. Specialized in surrogate modeling, digital twins, uncertainty quantification, and explainable predictive models for complex dynamic systems.

RESEARCH FOCUS

- Surrogate and reduced-order modeling for high-fidelity mechanical simulations.
- Multimodal data pipelines combining experiments, finite element simulations, and time-series signals.
- Explainable and uncertainty-aware AI models for friction-induced vibration and brake squeal.

RESEARCH EXPERIENCE

PhD Candidate

2023 – Present

LaMcube & LAMIH · Lille & Valenciennes, France

Thesis objective: predictive and explainable models for brake squeal using experimental and numerical data.

- Acquired and structured instrumented test data from multimodal sensors and numerical simulations.
- Developed surrogate models to accelerate prediction of the dynamic behavior of a braking system from finite element simulations, time signals, and multimodal experimental data.
- Built end-to-end workflows for preprocessing, model training, validation, uncertainty analysis, and explainability.
- Performed semantic segmentation of brake pad material images (SEM and optical), using unsupervised pseudo-labeling and transfer learning with Transformers.
- Authored 2 conference presentations (CSMA, CFM), 1 published article (MSSP), and 2 submitted articles (MSSP, JSV).

Teaching Assistant

2023 – Present

Polytech Lille & INSA Hauts-de-France · Lille & Valenciennes, France

- *Dynamics & signal processing* – taught system dynamics and fundamentals of signal processing.
- *MATLAB programming* – supervised projects in finite element simulation, optimization, and data analysis.

INDUSTRY EXPERIENCE

Simulation & Design Engineer – Apprentice

2020 – 2023

Alstom · Petite-Forêt, France

- Designed and validated railway products through mechanical simulation and data analysis.
- Collaborated with multidisciplinary and international teams across several sites.

Mechanical Engineering Intern

May – Aug. 2021

Canray · Bursa, Turkey

- Conducted test-simulation correlations on automotive components.
- Supported quality monitoring and technical exchanges with the client.

EDUCATION

PhD – Structural Dynamics & Scientific Machine Learning

2023 – Present

University of Lille · Lille, France

- *"Friction-induced vibrations: experiments and modeling of triggering conditions through enriched interface parameters."*
- Supervised by Prof. P. Dufrénoy and Prof. M. Massa.

Engineering Degree – Mechanical Simulation

2018 – 2023

INSA Hauts-de-France · Valenciennes, France

- Core coursework: optimization, finite element methods, programming, and dynamics.